The impact of punctuated infectiousness on infectious disease dynamics

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Abstract

1 Introduction

The trajectory of an epidemic is governed by two key factors: the basic reproduction number (R_0), which is the expected number of secondary infections that an infectious person will produce in an otherwise susceptible population; and the generation interval distribution, $g(\tau)$, which is the distribution of times at which those secondary infections are expected to occur relative to the index case's infection ($\tau = 0$) [Wallinga and Lipsitch, 2007].

Both the reproduction number and the generation interval distribution are expectations taken over a large population. At the individual level, the individual reproduction number (ν_i) is the expected number of infections that a particular infectious person i will produce, such that $R_0 = E_i[\nu_i]$ [Lloyd-Smith et al., 2005]. Similarly, the "individual infectiousness kernel" ($a_i(\tau)$) describes the time distribution of new infections caused by a particular index case i, such that $g(\tau) = E_i[a_i(\tau)]$ [Champredon and Dushoff, 2015, Svensson, 2007].

For many pathogens, the shape of the individual infectiousness kernel $a_i(\tau)$ is poorly understood. Contact tracing to count infections is hard enough; measuring precisely the times of those infections is even harder. When it is done, it is usually averaged over many individuals, so that the "infectiousness period" assigned to a person in models reflects some average over multiple individuals—either the generation interval itself, or some intermediate between the individual infectiousness kernel and the generation interval distribution. Some evidence suggests that the infectious period for SARS-CoV-2 and influenza, at the individual level, may be highly punctuated, such that opportunities for superspreading only exist over the course of a few hours.

At the individual level, communicating about the likely timing of a person's infectiousness is important: it is good to know if you will be very infectious for a short time, or mildly infectious for a long time, so that you can shape your behavior accordingly. Additionally, a punctuated individual infectiousness kernel could impact epidemic dynamics. Even if the epidemic has the same mean-field behavior, it could impact the stochastic dynamics via highly synchronized transmission events, especially when an outbreak is small. It could interplay with control measures, especially those designed to intervene on infectiousness at the individual level like test- or symptom-based screening and isolation. Indeed, the impact of the punctuatedness of the individual infectiousness kernel has been examined as a sensitivity analysis in various epidemiological studies. **Dan; others?** Stochastic effects are known to cascade through macroscopic features of an epidemic **Morris2025**. However, we lack a comprehensive understanding of how the sharpness of the individual infectiousness kernel, independent of other features like the individual reproduction number, basic reproduction number, and generation interval distribution, impacts epidemic dynamics under both uncontrolled and controlled settings. Furthermore, we lack an understanding of how to measure the shape of the individual infectiousness kernel reliably, along with the settings when the kernel's shape is statistically identifiable.

Here, we address these gaps.

2 The individual infectiousness profile

2.1 Expressing the Hudson–Kermack–McKendrick model in terms of individual infectiousness

Early work by Hudson, and later by Kermack and McKendrick [Kermack and McKendrick, 1927], modeled disease transmission in a population using an integral equation. In modern notation [Breda et al., 2012], the force of infection is

$$F(t) = \int_0^\infty F(t-\tau)\, S(t-\tau)\, A(\tau)\, d\tau \tag{1}$$

where $F(t)$ is the force of infection at time t, $S(t)$ is the density of susceptible individuals in the population at time t, and $A(\tau)$ is the population-level "infectiousness profile", a curve describing the expected contribution to the force of infection from an individual who was infected τ time units ago. The incidence of disease is simply the product of the force of infection and the susceptible density (i.e., $-\frac{dS}{dt} = F(t) \cdot S(t)$); thus, the infectiousness profile $A(\tau)$ is the fundamental object that determines how an epidemic unfolds.

The infectiousness profile is related to the basic reproduction number, R_0, and the generation interval, $g(\tau)$ [Breda et al., 2012]:

$$R_0 = \int_0^\infty A(\tau)\, d\tau \tag{2}$$

$$g(\tau) = \frac{A(\tau)}{R_0} \tag{3}$$

Furthermore, particular choices of $A(\tau)$ yield disease transmission dynamics that are equivalent to more familiar ordinary differential equation-based models [Breda et al., 2012]. For example, the SEIR model

$$\frac{dS}{dt} = -\beta S I \tag{4}$$

$$\frac{dE}{dt} = \beta S I - \gamma E \tag{5}$$

$$\frac{dI}{dt} = \gamma E - \alpha I \tag{6}$$

$$\frac{dR}{dt} = \alpha I \tag{7}$$

is obtained when

$$A(\tau) = \beta \frac{\gamma}{\gamma - \alpha} \left(e^{-\alpha\tau} - e^{-\gamma\tau} \right) \tag{8}$$

An individual person's infectiousness kernel $a_i(\tau)$ may differ substantially from $A(\tau)$, and different $a_i(\tau)$ may yield the same $A(\tau)$. For example, the SEIR model is often conceptually motivated by considering an individual who undergoes a latent period of length $X \sim \text{Exp}(\gamma)$ during which they are not infectious, followed by an infectious period of length $Y \sim \text{Exp}(\alpha)$ during which they

are infectious at level β. Mathematically,

$$a_i(\tau) = \begin{cases} \beta & \tau \in [\tau_i^{\text{on}}, \tau_i^{\text{on}} + \tau_i^{\text{off}}] \\ 0 & \text{otherwise} \end{cases} \quad \text{where } \tau_i^{\text{on}} \sim \text{Exp}(\gamma) \text{ and } \tau_i^{\text{off}} \sim \text{Exp}(\alpha) \tag{9}$$

It can be shown that $E_i[a_i(\tau)] = \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} a_i(\tau) = A(\tau)$ [Breda et al., 2012]. However, other $a_i(\tau)$ also yield the same $A(\tau)$ (**Figure 1**), including

$$a_i(\tau) = \beta \frac{\gamma}{\gamma - \alpha} \left(e^{-\alpha\tau} - e^{-\gamma\tau}\right) \tag{10}$$

where each person's infectiousness profile is a smooth curve that is identical to the population-level infectiousness profile, and

$$a_i(\tau) = \frac{\beta}{\gamma} \cdot \delta_{t_i}(\tau) \tag{11}$$

where $\delta_{t_i}(\tau)$ is a delta function with mass at time t_i, and t_i is distributed according to the generation interval density $A(\tau)/R_0$. In this case, a person's infectiousness is concentrated at a single moment. These choices of $a_i(\tau)$ yield the same mean-field epidemic dynamics, but their stochastic dynamics may differ in important ways.

Figure 1: The individual infectiousness profile impacts stochastic epidemic trajectories. (A-C) Schematic diagrams of three individual infectiousness profiles that yield equivalent mean-field dynamics: the smooth (A), stepwise (B), and spike (C) profiles (Eq xx, xx, xx). (D-I) Deterministic epidemic trajectories (blue) are numerical solutions of the mean-field renewal equation model, equivalent to a standard SEIR model with mean latent period XX and mean infectious period XX. For each individual infectiousness profile type, 1,000 stochastic simulations were generated (grey), depicting the cumulative infections (D-F) and daily new infection counts (G-I) for each profile type.

2.2 Properties and components of the individual infectiousness profile

Like $A(\tau)$, $a_i(\tau)$ is related to the individual reproduction number (ν_i) and the individual generation interval distribution ($\xi_i(\tau)$):

$$\nu_i = \int_0^\infty a_i(\tau)\, d\tau \tag{12}$$

$$\xi_i(\tau) = a_i(\tau)/\nu_i \tag{13}$$

Rearranging Eqs. (12)–(13), the individual infectiousness profile can be expressed as the product of the individual generation interval distribution and the individual reproduction number:

$$a_i(\tau) = \xi_i(\tau)\,\nu_i \tag{14}$$

However, a different and more general decomposition is also possible, splitting $a_i(\tau)$ into factors capturing a person's "biological potential", $b_i(\tau)$, and "contact potential", $c_i(t)$:

$$a_i(\tau) = b_i(\tau)\,c_i(t_i + \tau) \tag{15}$$

where t_i is person i's infection time, so τ represents time since infection. The biological potential $b_i(\tau)$ is a density that integrates to 1 ($\int_0^\infty b_i(\tau)\,d\tau = 1$) that describes how a person's infectiousness is spread over time. The contact potential, $c_i(t)$, is a nonnegative time-varying function that describes the expected number of individuals person i would infect if all their infectiousness were concentrated at the single moment t (i.e., if $b_i(\tau)$ were a delta function). Thus, $c_i(t)$ incorporates both a person's contact rate and the probability of transmission given contact. It can immediately be seen that if $c_i(t)$ is constant, then we recover Eq. (14) where $b_i(\tau) = \xi_i(\tau)$ (the individual generation interval distribution) and $c_i(t) = \nu_i$ (the individual reproduction number).

This decomposition feeds back into the population-level model. If we assume that b_i and c_i are independent (i.e., an individual's biological infectiousness is independent of their contact pattern), then

$$A(\tau) = E[a_i(\tau)] = E[b_i(\tau)]E[c_i(t_i + \tau)] = B(\tau) \cdot C(t_i + \tau) \tag{16}$$

where B is the population-averaged biological infectiousness potential and C is the population-averaged contact potential. Likewise, if C is constant in time, this reduces to the familiar

$$A(\tau) = g(\tau)\,R_0 \tag{17}$$

Splitting $a_i(\tau)$ into biological potential b_i and contact potential c_i allows us to separately consider how the trajectory of a person's infectiousness, unconditional on their contact patterns, impacts transmission dynamics under various contact patterns.

2.3 A Gamma convolutional model for the individual infectiousness profile

The "stepwise" expression for $a_i(\tau)$ in Eq. (9) differs from the "smooth" and "spike" expressions in Eqs. (10)–(11) in a fundamental way: For Eq. (9), ν_i differs across individuals ($\nu_i \sim$ XX), while for Eqs. (10)–(11), ν_i is uniform across individuals ($\nu_i = R_0$). As a result, the probability of epidemic extinction differs for the stepwise profile relative to both the smooth and spike profiles – but this is due to an inherent difference in the distribution of the individual reproduction number, not to the timing of the infections. Because of this fundamental dissimilarity—and because the impact of variation in ν_i on epidemic dynamics has been discussed extensively elsewhere [Lloyd-Smith et al., 2005]—we will focus most of the manuscript on individual infectiousness profiles of the type

in Eqs. (10)–(11), where the individual infectiousness ν_i is uniform across the population. That way, the impact of the timing of infectiousness can be examined in isolation.

To achieve this, we seek a family of functions that allow us to model individual infectiousness profiles with the following characteristics:

- All have uniform total infectiousness: $\int_0^\infty a_i(\tau)\, d\tau = R_0$
- Smooth, single-peaked distributions
- All have identical shape, except for a time shift (no mixtures of broad smooth distributions with narrow spiky distributions)

A candidate set of functions that obeys this is the set of Gamma convolutional curves. Let $A(\tau)$ be Gamma distributed (up to a scaling constant):

$$A(\tau) = R_0\, f_A(\tau) \qquad \text{where } f_A(\tau) \sim \text{Gamma}(\alpha, \beta), \quad \beta = \alpha/\mu \tag{18}$$

where μ is the mean generation interval and α controls the shape of the population-level infectiousness profile.

Now, let

$$a_i(\tau) = R_0\, f_a(\tau - l_i) \qquad \text{where } f_a(\tau) \sim \text{Gamma}(\psi\alpha, \beta) \quad \text{and} \quad l_i \sim \text{Gamma}((1-\psi)\alpha, \beta) \tag{19}$$

Here, $\psi \in (0, 1)$ is a parameter governing the punctuatedness of $a_i(\tau)$, with smaller ψ yielding a more concentrated infectiousness profile (**Figure XX**). At the limits, $\psi \to 0$ yields a delta function with mass at time-since-infection $\tau = l_i$, and $\psi \to 1$ yields an individual infectiousness profile identical to the population-level profile. Since the sum of two Gamma-distributed random variables with the same rate is also Gamma distributed (with the same rate and with shape equal to the sum of the two component shapes), this formulation is guaranteed to converge to $A(\tau)$ in expectation (**Supplementary Methods**). Furthermore, the ψ-values may differ across individuals; regardless of the distribution they are drawn from, the Gamma sum property ensures that $E_i[a_i(\tau)] = A(\tau)$ (**Supplementary Methods**, **Figure XX**).

2.4 Stochastic simulations

To simulate the infections produced by a single person, we generate the number of infections using a Poisson(ν_i) distribution and distribute them according to $a_i(\tau)/\nu_i$. For the Gamma convolutional model, the timing distribution has closed form and is easy to sample from using built-in packages. For more arbitrary individual infectiousness profiles (e.g., obtained by convolving a Gamma-distributed $b_i(\tau)$ with a periodic contact function $c_i(t)$), we use Poisson thinning:

3 The impact of the individual infectiousness profile on uncontrolled epidemics

3.1 Punctuated individual infectiousness profiles generate later and more variably-timed epidemics.

Even in large, well-mixed populations, stochasticity can durably impact the trajectory of an epidemic. One way this stochasticity enters is via a random time shift, where epidemics take a variable amount of time to reach the exponential growth phase [Morris et al., 2024], leading to variation in the timing of the entire epidemic trajectory. Here, we address the question of how this time shift distribution depends on the shape of the individual infectiousness profile, holding all else equal.

Simulations confirm that narrower infectiousness profiles ($\psi \to 0$) yield more variable and later-shifted epidemic timings (**Figure XX**, **Supplementary Table Sxx**). Across 10,000 simulated epidemics in a population of size 1000 using SARS-CoV-2 omicron-like parameters, the median time to reach 100 infections was XX days for $\psi = 0$ (IQR: [XX, XX]) vs. XX days for $\psi = 1$ (IQR: [XX, XX]). The impact of ψ on epidemic timing was lessened with lower R_0 (influenza-like parameters, median time to 100 infections: XX days for $\psi = 0$ [IQR: XX, XX] vs. XX days for $\psi = 1$ [IQR: XX, XX]) and increased with higher R_0 (measles-like parameters, median time to 100 infections: XX days for $\psi = 0$ [IQR: XX, XX] vs. XX days for $\psi = 1$ [IQR: XX, XX]). From another perspective: by day XX of the simulated epidemic under measles-like parameters, XX% of the outbreaks with $\psi = 1$ had reached 100 infections, while only XX% of the outbreaks with $\psi = 0$ had reached 100 infections (**Figure XX**).

These observations are confirmed by the underlying theory. The cumulative incidence over time, $Z(t)$, during the early exponential growth phase of an epidemic can be approximated as a deterministic exponential growth function with a random initial condition scaling factor W:

$$Z(t) \approx W \cdot ce^{rt} \tag{20}$$

with exponential growh rate r, and where c is the mean-field initial condition (when $W = 1$).

With a change of variables, the cumulative incidence can be expressed as the same deterministic growht function with a random time shift ς:

$$Z(t) \approx ce^{r(t+\varsigma)} \tag{21}$$

where

$$\varsigma = \frac{\log W - \log E[W]}{r} \tag{22}$$

Here, $E[W]$ is the initial condition and r is the exponential growth rate, so $\varsigma \propto \log[W]$.

First, we consider the variance of the time shift, $Var[\varsigma]$. For ease of notation, we focus on $Var[W]$, noting that $Var[\varsigma]$ is monotonically related to $Var[W]$. It can be shown that **Supplementary Text**

$$Var[W] = m_2 + R_0^2 \rho^{2\psi\alpha}[\rho_2^{(1-\psi)\alpha} - \rho^{2(1-\psi)\alpha}]$$

which is maximized when $\psi = 0$ (the infectiousness profile is narrow) and becomes 0 when $\psi = 1$ (the infectiousness profile is maximally broad). (Can we say something about how this suggests that the variance of the time shift scales roughly linearly with ψ?)

Next, we consider the typical time shift, which we obtain by studying the mode of the distribution of W.

We can approximate the distribution of W using moment matching to a generalized Gamma distribution. With the Gamma-shaped individual infectiousness profiles, the moments have closed-form expressions. Using two-moment matching to a Gamma distribution, we find that

$$\text{mode}_\varsigma = \frac{1}{r}\log(1 - \text{CV}^2_{\text{cond}})]$$

Here, (description). Again, we see that the mode shifts later as $\psi \to 0$.

The theory aligns well with the simulations. Using generalized gamma distributions fit using moment matching to the first five moments of W, we get similar survival curves for the time to 100 infected (**Figure XX**).

3.2 Narrower individual infectiousness profiles yield greater uncertainty in epidemic growth rate estimates.

Based on simulations.

3.3 Narrower biological infectiousness profiles interact with periodic contact patterns to generate overdispersion in secondary infections.

Closed-form solution:

$$k = \frac{2}{\bar{c}^2} A^2 \Big(1 + \frac{\omega^2}{r^2}\Big)^{\psi\alpha} = \frac{2}{\bar{c}^2} A^2 \Big(1 + (\frac{z^*_{per}}{z_{per}})^2\Big)^{\psi\alpha}$$

Figure 2: Impact of the individual infectiousness profile on epidemic timing. Survival curves for time-to-100. growth rate estimate box-and-whiskers. Overdispersion/k vs psi heatmap.

4 The impact of the individual infectiousness profile on detect-and-isolate interventions

- The biologically anchored detection model
- For fixed detection offset, TE depends on psi. Skewed profiles buy a bit of detection time after the peak where TE is still greater for spikier profiles (within reason).
- Symptoms arising somewhat randomly smooth out the effect. For spikier profiles, they boost TE when mean detection is after the peak infectiousness and reduce TE when mean detection is beffore peak infectiousness – meaningfully.
- Regular screening: TE is sensitive to screening frequency and delays in response, especially in specific regions of sensitivitiy-frequency-window size combinations.
- TE is only part of the story: isolation shrinks generation intervals (moreso for smooth profiles), and increases overdispersion (moreso for spike profiles), which has an impact on epidemic dynamics that differ from what you'd expect if you simply ran an epidemic with $R_{\text{eff}} = R_0(1 - TE)$. Give some simulations to demonstrate.

5 The impact of individual infectiousness on gathering size restrictions

Briefly, we also consider epidemic control by restricting gathering sizes.

- The theory
- The model
- Like with detect-and-isolate, the reduction in R_{eff} isn't the whole story: gathering size restrictions impact overdispersion, and again do so more for spike than for smooth profiles.

6 Notes on detect and isolate

We first consider the detect-and-isolate interventions. Detection may occur through various mechanisms, e.g., the onset of symptoms or a positive diagnostic test.

We can express the effectiveness of a detect-and-isolate intervention in terms of the *test effectiveness* [Middleton and Larremore, 2024].

If the detection mechanism is independent of the individual infectiousness profile (e.g., detection is anchored to the infection time but does not depend on what happens thereafter), the imapct of the individual infectiousness profile washes out (**Supplementary Text**). However, if detectability is anchored somehow to the infectiousness profile – which is likely to be the case in general – the individual profile begins to matter.

Here, we consider detection strategies that are anchored to the mode of the individual infectiousness profile (i.e., the time of peak infectiousness). Let $D(\tau)$ be the cumulative probability that a person has been detected by time τ.

To begin, we consider a fixed, deterministic detection time relative to the mean - this might correspond to symptoms arriving deterministically some amount of time prior to (or after) peak infectiousness.

Maybe: just show a TE profile for a spike, smooth, and $\psi = 0.5$ profile for a deterministic offset (size of offset on the horizontal axis, TE on the vertical); and then show the analogous curves for when the detection profile is random (say, normal, where now the horizontal axis captures the mean of the distribution, and we pick some reasonable standard deviation). This should illustrate how the timing of detection (with variation) interplays with the punctuation of the infectiousness profile. That might be enough here.

Then we can focus on the implications for the generation interval distribution and secondary infection distribution.

I want to try my hand at making this figure: what do we need?

- The general detection distribution framework
- Narrower infectiousness profiles transfer detection into an all-or-nothing proposition
- Detection and isolation shorten the generation interval distribution, moreso for smoooth profiles, which means that epidemics may grow (and shrink) faster than a naive application of TE would suggest...
- while detection and isolation increases overdispersion in secondary infections, moreso for spike profiles, so that epidemics are less likely to take off (but more explosive) than would be expected with a naive application of TE.
- The takehome: detection and isolation reduce R_{eff} via TE, but a simple application of TE won't tell us everything we need to know: instead, we need to know something about the individual infectiousness profile to properly anticipate how detect-and-isolate interventions will impact epidemic dynamics.
- Furthermore: narrower infectiousness profiles make detect-and-isolate interventions sensitive to small shifts, especially near the infectiousnesss profile itself. So, if symptoms move from slightly after to slighlty before infectiousness on average, this will have an outsize impact on narrow profiles vs. smooth ones. Similarly, if a small increase in testing cadence could have a huge impact for spike profiles, if we shift the mean of D from slightly after to slightly before peak infectiousness; but in other areas, it'll hardly have any impact on spike profiles, whereas the change in effectiveness for smooth profiles should be more gradual.
- This last point is relevant for when detection is anchored to the mode (or otherwise anchored to the individual infectiousness profile); otherwise, using the population average I think should be fine (e.g. if detection depends only on time of infection but not subsequent infectiousness).

- The takeaway from this: if we want to know the impact of shifting biology (changes in the symptom timing) or shifts in recommendations (changes in the cadence of testing), we should know something about the individual infectiousness profile (and how it relates to detectability).
- Adding in delays/turnaround time will also impact all this.

7 Identifiability and inference of the individual infectiousness profile's shape

While the concentration of the individual infectiousness profile can impact how an epidemic unfolds, we lack frameworks for estimating it. Many approaches exist for estimating the generation interval, but there has not been a systematic approach developed to assess how individual infectiousness varies and to specify where on the continuum we sit—i.e., how much of the variation in the generation interval distribution is attributable to variation between individuals, and how much to variation in infection timing within individuals.

We begin by assessing under which conditions we can hope to infer the shape of the individual infectiousness profile. We restrict to gamma-distributed timings, and consider cases where we assume that κ is shared across individuals and where κ comes from a beta distribution governed by hyperparameters α and β. Then, when κ is identifiable, we introduce a hierarchical Bayesian inference strategy to infer the shape of the individual infectiousness profile given contact tracing data.

For this section: let's imagine that there's a single psi value shaed among everyone. We can use a Beta prior. Two things to consider: how do we estimate ψ given a list of infection times (assume we know them exactly)? And how do we estimate ψ given infection times and knowledge of $c_i(t)$? My hope is that this will emphasize that knowing the "negatives" – i.e., the contacts that "should" have led to transmission but did not – improves our inferential power substantially. Then, I want to think about how to achieve a given power in each scenario across R0, assuming some fixed generation interval distribution. That should be enough for now. We should probably include some amount of uncertainty in infection times for this base analysis, and we should post code for running the analysis given a line list of infection times and contacts. In the Discussion, we need to address the fact that there could be different ψ_i values across individuals, themselves drawn from a distribution, so we'd want some kind of hierarhcical Bayesian model for inference here; also, we don't consider differences in the height (i.e. inherent biological differences in ν_i / b_i), which is another key thing to include in the inference.

8 Discussion

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Supplementary Information

Supplementary Methods

Supplementary Figures

Supplementary Tables